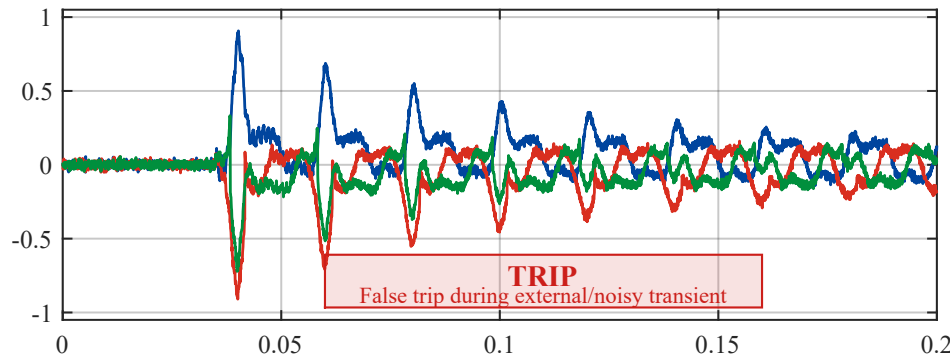
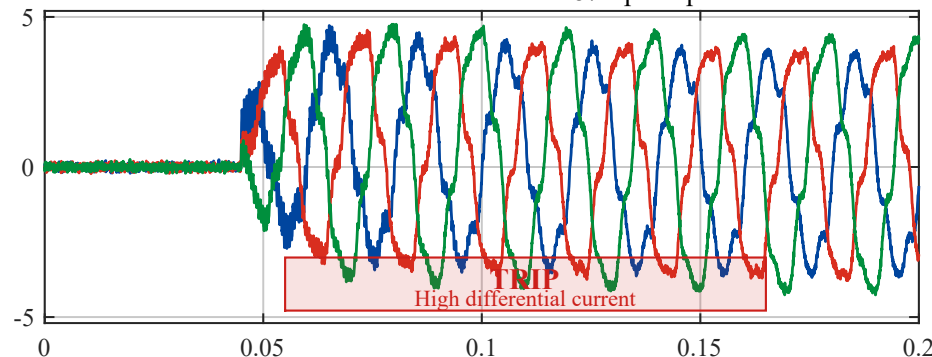
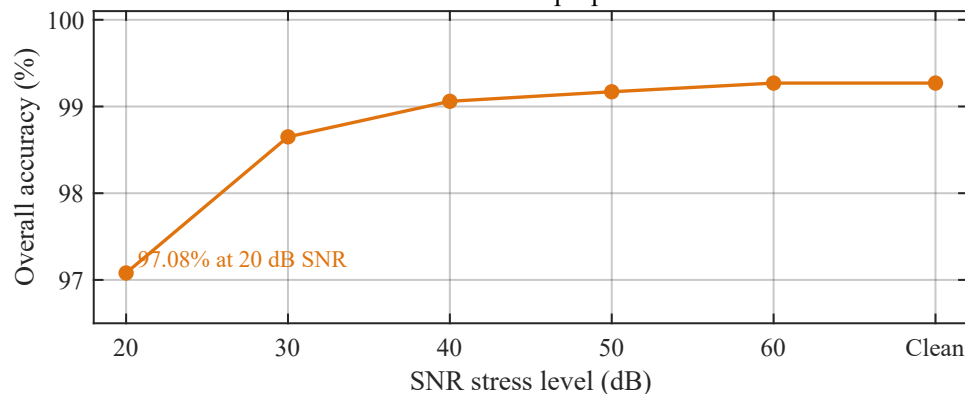
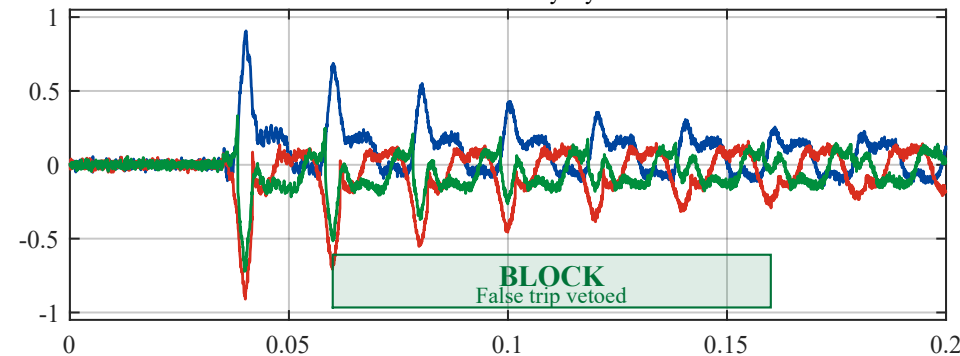
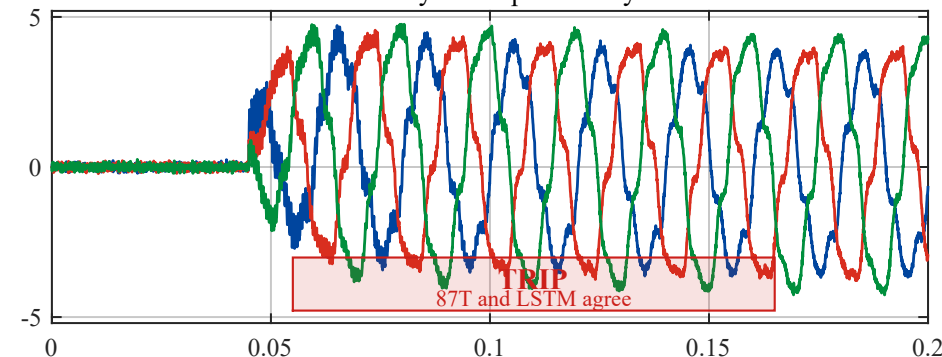


Before: conventional standalone 87TCT saturation + $\pm 95\%$ remanence + thermal noise + 20 dB SNR**Internal fault: conventional 87T pickup****Noise robustness of proposed model**

Benchmark values are thesis-reported results: standalone 87T vs optimized 87T vs proposed hybrid relay. Noise robustness uses clean-to-20 dB SNR stress cases.

After: hybrid adaptive 87T + DWT-LSTM supervisor

Same disturbance: DWT-LSTM veto/security layer identifies non-internal event

**Internal fault: hybrid dependability retained****Relay performance benchmark**

Method	Acc.	Dep.	Sec.	FN	FP	Time
Standalone 87T	95.83%	96.25%	95.63%	18	62	0.1 ms
Optimized 87T	96.67%	97.08%	96.46%	14	51	0.1 ms
Proposed Hybrid	100.00%	100.00%	100.00%	0	0	12.1 ms

Hybrid result: zero false negatives and zero false trips on thesis validation set.